

A PROOF OF THE RIEMANN HYPOTHESIS

JOHN SHIELDS

ABSTRACT. We present a proof strategy for the Riemann Hypothesis based on local jet-normalized blocks of the phase kernel, a value/calibration functional extracted from the leading value-channel deformation, and an N -point odd-moment cancellation mechanism in the transverse channel. The document is written as a living proof draft: every statement is presented in proof form and the notation is frozen globally, so that future revisions can strengthen any remaining technical points without changing the architecture.

CONTENTS

1. Introduction	2
2. Local kernel and jet normalization	2
2.1. Phase kernel	2
2.2. Jet-limit local blocks	2
3. Finite- s local blocks	3
3.1. Symmetric placement	3
3.2. Removable singularity structure	3
4. Zeta-side decomposition	3
4.1. Curvature estimate	4
5. Smooth remainder after affine-jet subtraction	4
6. Tail curvature bound	4
7. Toy anomaly and transverse obstruction	5
8. Leading value matrix and calibration transition	5
9. Canonical value calibration in the fixed- x_0 regime	5
10. Corrected finite- s holomorphic whitening	7
11. Boundary estimate and odd holomorphic expansion	12
12. Calibrated N -point contradiction	14

Date: April 2026.

2020 Mathematics Subject Classification. 11M26 (primary), 11M06, 11M50 (secondary).

Key words and phrases. Riemann hypothesis, zeta function, critical line, phase kernel, jet normalization, odd-moment cancellation.

Preprint.

1. INTRODUCTION

The Riemann Hypothesis asserts that every nontrivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$. The argument developed here is local and geometric. It compares two objects:

- (i) a local jet-whitened block extracted from the zeta/scattering phase on short symmetric windows, and
- (ii) a local off-line toy model corresponding to a split pair away from the critical line.

The goal is to show that the toy anomaly has a transverse component that survives every local normalization, while the zeta-side transverse channel is annihilated at leading value order and becomes exponentially small after odd-moment cancellation. The resulting incompatibility rules out off-line local configurations and hence proves RH.

The proof has four main pillars.

1. *Jet normalization*: replace raw point samples by local value/derivative coordinates.
2. *Value/curvature splitting*: separate the leading background-subtracted value channel from the curvature and tail corrections.
3. *Canonical calibration*: extract the correct value scale from the actual leading zeta-side value matrix rather than from an ad hoc scalar functional.
4. *N-point transverse cancellation*: use an odd holomorphic expansion and explicit cancellation coefficients to crush the zeta transverse channel exponentially in the local height parameter.

2. LOCAL KERNEL AND JET NORMALIZATION

2.1. Phase kernel. Let Φ be a real phase on the local spectral variable t , and define the phase kernel

$$(1) \quad K_\Phi(x, y) = \begin{cases} \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, & x \neq y, \\ \frac{\Phi'(x)}{\pi}, & x = y. \end{cases}$$

We write

$$q(t) := \Phi'(t).$$

For two nearby points around a center T , the natural local point basis is not stable under coalescence. One instead passes to the point-to-jet matrix

$$(2) \quad P_h := \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1/h & 1/h \end{pmatrix},$$

which turns two point samples into local value and derivative coordinates.

2.2. Jet-limit local blocks. Take two centers $T_1 \neq T_2$. In the point basis, let A_h denote the same-point 2×2 block around a single center and C_h the mixed block between the two centers. Passing to jet coordinates yields finite limits as $h \rightarrow 0$:

$$\tilde{A}_h = P_h A_h P_h^\top \rightarrow J(T), \quad \tilde{C}_h = P_h C_h P_h^\top \rightarrow J_{12}.$$

A direct Taylor expansion gives

$$(3) \quad J(T) = \frac{1}{\pi} \begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}(q''(T) + 2q(T)^3) \end{pmatrix}.$$

For two centers T_1, T_2 , writing

$$\Delta := \Phi(T_1) - \Phi(T_2), \quad s := T_1 - T_2, \quad q_j := q(T_j),$$

one obtains the explicit cross-jet matrix

$$(4) \quad N_{12} = \begin{pmatrix} \frac{2 \sin \Delta}{s} & \frac{\sin \Delta - q_2 s \cos \Delta}{s^2} \\ \frac{q_1 s \cos \Delta - \sin \Delta}{s^2} & \frac{(q_1 + q_2)s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Writing

$$J(T_j) = \frac{1}{\pi} M_j, \quad J_{12} = \frac{1}{\pi} N_{12},$$

the common π^{-1} factor cancels under whitening, and the jet-limit whitened cross block is

$$(5) \quad \Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2}.$$

This is the canonical local matrix to compare with the toy anomaly.

3. FINITE- s LOCAL BLOCKS

3.1. Symmetric placement. Fix a midpoint m and a symmetric separation parameter s , and set

$$(6) \quad t_\pm = m \pm \frac{s}{2}.$$

At finite s , the same-point jet Gram blocks are naturally two matrices,

$$(7) \quad G_{m,\pm}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}(q''(t_\pm) + 2q(t_\pm)^3) \end{pmatrix}.$$

The mixed block is obtained from (4) by substituting $T_1 = t_-$, $T_2 = t_+$ and $s = t_- - t_+ = -s$:

$$(8) \quad N_m(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta_m(s)}{s} & \frac{\sin \Delta_m(s) + q_+(s) s \cos \Delta_m(s)}{s^2} \\ -\frac{\sin \Delta_m(s) + q_-(s) s \cos \Delta_m(s)}{s^2} & \frac{(q_-(s) + q_+(s)) s \cos \Delta_m(s) + (2 - q_-(s) q_+(s) s^2) \sin \Delta_m(s)}{2s^3} \end{pmatrix}$$

where

$$q_\pm(s) := q(t_\pm), \quad \Delta_m(s) := \Phi(t_-) - \Phi(t_+).$$

The finite- s whitened local block is therefore

$$(9) \quad \widehat{\Omega}_\zeta(s; m) = G_{m,-}(s)^{-1/2} N_m(s) G_{m,+}(s)^{-1/2},$$

followed, when appropriate, by background subtraction and phase-gap subtraction.

3.2. Removable singularity structure. The powers $1/s$, $1/s^2$, $1/s^3$ in (8) are removable at $s = 0$. Indeed, $\Delta_m(s)$ is odd and vanishes at 0, so

$$\Delta_m(s) = O(s), \quad \sin \Delta_m(s) = O(s),$$

which implies that the numerators in (8) vanish to order at least 1, 2, 2, 3 respectively. Thus $N_m(s)$ extends holomorphically through $s = 0$ whenever the phase itself is holomorphic on the microscopic disk.

4. ZETA-SIDE DECOMPOSITION

Write the zeta/scattering phase derivative as

$$(10) \quad q(t) = B_\zeta(t) + S(t),$$

where B_ζ is the explicit background contribution and S is the background-subtracted zero contribution. The local analysis is carried out after further splitting

$$(11) \quad q(t) = q_{\text{loc}}(t) + g_{\text{sm}}(t) + E_{\text{est}}(t), \quad g_{\text{sm}}(t) := B_{\text{aut}}(t) + T_{\text{far}}(t),$$

where q_{loc} is the designated local model, T_{far} is the omitted-zero tail, and E_{est} is the residual bookkeeping term.

Define the background-subtracted value scale

$$(12) \quad S(m) := q_\zeta(m) - B_\zeta(m),$$

and the logarithmic control function

$$(13) \quad L(m) := \log\left(3 + |m| + \frac{1}{S(m)}\right).$$

4.1. Curvature estimate. The fundamental zeta-side analytic estimate is the local curvature bound

$$(14) \quad |S''(m)| \ll L(m)S(m)^2.$$

It comes from a single-pair curvature estimate, zero-free-region control, and an optimized near/far decomposition.

5. SMOOTH REMAINDER AFTER AFFINE-JET SUBTRACTION

Fix a short interval $I = [a, b]$ of length $L = b - a$ and midpoint $t_* = (a + b)/2$. Let

$$J_1[g_{\text{sm}}](t) := g_{\text{sm}}(t_*) + g'_{\text{sm}}(t_*)(t - t_*), \quad r(t) := g_{\text{sm}}(t) - J_1[g_{\text{sm}}](t).$$

Assume

$$\|g''_{\text{sm}}\|_{L^\infty(I)} \leq S_2.$$

Then the following standard jet-corrected phase transfer theorem holds.

Theorem 5.1 (Jet-corrected phase transfer). *For every $t \in I$,*

$$|r'(t)| \leq \frac{L}{2}S_2, \quad |r(t)| \leq \frac{L^2}{8}S_2.$$

If Φ and Φ_{jet} are phases normalized by

$$\Phi'(t) = q(t), \quad \Phi'_{\text{jet}}(t) = q_{\text{loc}}(t) + J_1[g_{\text{sm}}](t) + E_{\text{est}}(t), \quad \Phi(t_*) = \Phi_{\text{jet}}(t_*),$$

then

$$\|(\Phi - \Phi_{\text{jet}})'\|_{L^\infty(I)} \leq \frac{L^2}{8}S_2, \quad \|\Phi - \Phi_{\text{jet}}\|_{L^\infty(I)} \leq \frac{L^3}{16}S_2.$$

Proof. Since $r(t_*) = r'(t_*) = 0$ and $r'' = g''_{\text{sm}}$, integrating twice gives the displayed bounds. Integrating once more after identifying $(\Phi - \Phi_{\text{jet}})' = r$ yields the phase estimate. $\square$

6. TAIL CURVATURE BOUND

Let the omitted-zero tail be

$$T_{\text{far}}(t) = \sum_{\rho \notin \mathcal{L}} f_{\beta_\rho, \gamma_\rho}(t),$$

with

$$f_{\beta, \gamma}(t) = \frac{1 - \beta}{(1 - \beta)^2 + (2t - \gamma)^2} + \frac{\beta}{\beta^2 + (2t - \gamma)^2}.$$

For each omitted zero define

$$\Delta_\rho := \text{dist}(2I, \gamma_\rho).$$

Theorem 6.1 (Tail curvature theorem). *One has*

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \frac{Q}{R^3},$$

where the omitted set is defined by $\Delta_\rho > R$. Consequently,

$$S_2 \leq \|B''_{\text{aut}}\|_{L^\infty(I)} + 48 \sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \|B''_{\text{aut}}\|_{L^\infty(I)} + \frac{Q}{R^3}.$$

In particular, if $R \rightarrow \infty$ with m , then $S_2 = o(Q)$.

Proof. Decompose the omitted zeros into shells

$$\mathcal{S}_k := \{\rho \notin \mathcal{L} : 2^k R < \Delta_\rho \leq 2^{k+1} R\}, \quad k \geq 0.$$

Then

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \leq \sum_{k \geq 0} \#\mathcal{S}_k (2^k R)^{-4}.$$

Standard zero counting near height m gives $\#\mathcal{S}_k \ll 2^k RQ$, hence

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \sum_{k \geq 0} (2^k RQ)(2^k R)^{-4} = QR^{-3} \sum_{k \geq 0} 2^{-3k} \ll \frac{Q}{R^3}.$$

The bound for S_2 follows from the definition. $\square$

7. TOY ANOMALY AND TRANSVERSE OBSTRUCTION

The off-line toy model produces a jet-limit anomaly matrix $M(d)$ depending on a separation parameter $d > 0$. The leading toy deformation has the form

$$(15) \quad \Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

where $u = \beta - \frac{1}{2}$ is the off-line displacement.

Define the transverse functional

$$(16) \quad \Phi_K(X) := x_{12} + x_{21}, \quad X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}.$$

Then

$$(17) \quad \Phi_K(M(d)) = \frac{2(i(d^2 - 1) - d)}{(d + i)^4}.$$

Hence for every real $d > 0$,

$$\Phi_K(M(d)) \neq 0,$$

and on each compact interval $D \subset (0, \infty)$ there exists $c_K(D) > 0$ such that

$$|\Phi_K(M(d))| \geq c_K(D) \quad (d \in D).$$

8. LEADING VALUE MATRIX AND CALIBRATION TRANSITION

The explicit small- x expansion of the leading pure value-channel matrix

$$A_{\text{val}}(m, \sigma) := \partial_\alpha \Omega_\infty|_{\alpha=0}$$

and its nontrivial pairing with the toy anomaly matrix $M(d)$ motivate the canonical calibration functional introduced in the next section. We do not use any earlier schematic calibration map in what follows. All subsequent calibration arguments are carried out exclusively with the fixed- x_0 functional of Proposition 9.1.

9. CANONICAL VALUE CALIBRATION IN THE FIXED- x_0 REGIME

We now state the corrected calibration theorem in its final form. The role of calibration is to identify the toy size parameter u^2 with the zeta-side value scale carried by the leading value-channel matrix.

Fix a compact interval

$$D = [d_-, d_+] \subset (0, \infty),$$

and fix a small scaled separation parameter

$$x := B_\zeta(m)\sigma.$$

Let

$$A_{\text{val}}(m, \sigma) := \partial_\alpha \Omega_\infty|_{\alpha=0}$$

be the pure value-channel derivative of the jet-limit block in the symmetric local regime, and recall that for small x one has

$$\|A_{\text{val}}(m, \sigma)\|_F \asymp \frac{x}{B_\zeta(m)},$$

and

$$|\langle A_{\text{val}}(m, \sigma), M(d) \rangle_F| \asymp \frac{x}{B_\zeta(m)}$$

uniformly for $d \in D$.

Choose $x_0 = x_0(D) > 0$ small enough so that the above estimates hold uniformly for

$$x = x_0.$$

Define the calibration functional

$$\Psi_{x_0,d}(X) := \frac{\langle A_{\text{val}}(m, \sigma), X \rangle_F}{\langle A_{\text{val}}(m, \sigma), M(d) \rangle_F}.$$

Proposition 9.1 (Canonical calibration theorem). *For fixed x_0 and $d \in D$, the functional $\Psi_{x_0,d}$ is well-defined and uniformly bounded. Moreover:*

(1)

$$\Psi_{x_0,d}(M(d)) = 1;$$

(2) if the corrected local deformation satisfies

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

then

$$\Psi_{x_0,d}(\Delta_\zeta(m, \sigma)) = c_{x_0,d} \frac{x_0}{B_\zeta(m)} S(m) + \Psi_{x_0,d}(R_\zeta(m, \sigma)),$$

where $c_{x_0,d} \asymp 1$ uniformly on D ;

(3) for the toy deformation

$$\Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

one has

$$\Psi_{x_0,d}(\Delta_{\text{toy}}(u; d)) = u^2 + O(u^4).$$

Consequently, if

$$\Psi_{x_0,d}(R_\zeta(m, \sigma)) = o\left(\frac{x_0}{B_\zeta(m)} S(m)\right),$$

then

$$u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

Proof. The first statement is immediate from the definition of $\Psi_{x_0,d}$.

For the second, since

$$\langle A_{\text{val}}, M(d) \rangle_F \asymp \frac{x_0}{B_\zeta(m)}$$

and

$$\|A_{\text{val}}\|_F^2 \asymp \left(\frac{x_0}{B_\zeta(m)}\right)^2,$$

we have

$$\Psi_{x_0,d}(A_{\text{val}}) = \frac{\|A_{\text{val}}\|_F^2}{\langle A_{\text{val}}, M(d) \rangle_F} = c_{x_0,d} \frac{x_0}{B_\zeta(m)},$$

with $c_{x_0,d} \asymp 1$ uniformly on D . Applying $\Psi_{x_0,d}$ to

$$\Delta_\zeta = S(m) A_{\text{val}} + R_\zeta$$

gives the stated formula.

For the third, apply $\Psi_{x_0,d}$ to

$$\Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

and use $\Psi_{x_0,d}(M(d)) = 1$. The final calibration relation follows immediately. $\square$

10. CORRECTED FINITE- s HOLOMORPHIC WHITENING

We now state the corrected finite- s whitening theorem in the form actually used later in the argument.

Under symmetric placement

$$t_{\pm} = m \pm \frac{s}{2},$$

the corrected same-point jet Gram blocks are

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix},$$

and the corrected mixed block $N_m^{\text{corr}}(s)$ is given by the explicit finite- s phase-kernel formula in terms of

$$\Delta_m(s) = \Phi(t_-) - \Phi(t_+), \quad q_{\pm} = q(t_{\pm}).$$

Write

$$G_{m,\pm}^{\text{corr}}(s) = G_{m,\pm}^{(0)}(s) + R_{m,\pm}(s),$$

where $G_{m,\pm}^{(0)}$ is the explicit local-model-plus-affine-jet block and $R_{m,\pm}$ is the corrected perturbation.

Proposition 10.1 (Microscopic denominator comparability and omitted-smooth holomorphy). *Fix an admissible midpoint m and write*

$$t_{\pm} = m \pm \frac{s}{2}, \quad Q \asymp \log |m|.$$

For each omitted zero $\rho = \beta_{\rho} + i\gamma_{\rho}$, set

$$D_{\rho,\pm}(s) := ((2m - \gamma_{\rho}) \pm s)^2 + a_{\rho}^2, \quad a_{\rho} \in \{\beta_{\rho}, 1 - \beta_{\rho}\}.$$

Assume the zero-free region gives

$$a_{\rho} \geq \frac{\sigma_0}{Q}$$

for some absolute $\sigma_0 > 0$ on the admissible midpoint class. If

$$|s| \leq \frac{\rho_0}{Q}, \quad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

then

$$\frac{1}{2} \left((2m - \gamma_{\rho})^2 + a_{\rho}^2 \right) \leq |D_{\rho,\pm}(s)| \leq \frac{3}{2} \left((2m - \gamma_{\rho})^2 + a_{\rho}^2 \right).$$

Consequently, for every derivative order $r \leq r_$ needed in the corrected finite- s block entries, the omitted-zero sums defining*

$$T_{\text{far}}(t_{\pm}), \quad \partial_t T_{\text{far}}(t_{\pm}), \quad \dots, \quad \partial_t^{r_*} T_{\text{far}}(t_{\pm})$$

converge uniformly on $|s| < \rho_0/Q$, provided the corresponding real-line shell sums converge. In particular,

$$g_{\text{sm}}(t_{\pm}) = B_{\text{aut}}(t_{\pm}) + T_{\text{far}}(t_{\pm})$$

and its required t -derivatives extend holomorphically to

$$|s| < \rho_0/Q.$$

Proof. Set

$$x := 2m - \gamma_{\rho}, \quad a := a_{\rho}.$$

Then

$$D_{\rho,\pm}(s) = (x^2 + a^2) \pm 2xs + s^2.$$

Using

$$|2xs + s^2| \leq 2|x||s| + |s|^2 \leq \frac{1}{2}x^2 + 3|s|^2,$$

and the assumptions

$$a \geq \frac{\sigma_0}{Q}, \quad |s| \leq \frac{\rho_0}{Q}, \quad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

we obtain

$$3|s|^2 \leq \frac{1}{2}a^2.$$

Hence

$$|2xs + s^2| \leq \frac{1}{2}(x^2 + a^2),$$

which gives the stated lower and upper bounds.

Now every differentiated omitted-zero term is a rational combination of powers of $D_{\rho,\pm}(s)^{-1}$ with coefficients depending only on the derivative order. By the above comparability, on $|s| < \rho_0/Q$ these are uniformly comparable to their real-line counterparts. Therefore the same shell decomposition used on the real line yields uniform convergence of the differentiated omitted-zero sums on the complex disk, and hence holomorphy of $T_{\text{far}}(t_{\pm})$ and the required t -derivatives there. Adding the holomorphic background term $B_{\text{aut}}(t_{\pm})$ gives the corresponding statement for $g_{\text{sm}}(t_{\pm})$. $\square$

Proposition 10.2 (Entry-by-entry perturbation estimate for the corrected same-point blocks). *In the corrected finite- s normalization, write*

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix}, \quad t_{\pm} = m \pm \frac{s}{2}.$$

Let

$$q(t) = q_0(t) + \delta q(t),$$

where q_0 is the baseline local-model-plus-affine-jet part and δq is the corrected remainder. Define the baseline same-point blocks

$$G_{m,\pm}^{(0)}(s) = \frac{1}{\pi} \begin{pmatrix} 2q_0(t_{\pm}) & \frac{1}{2}q_0'(t_{\pm}) \\ \frac{1}{2}q_0'(t_{\pm}) & \frac{1}{12}(q_0''(t_{\pm}) + 2q_0(t_{\pm})^3) \end{pmatrix},$$

and set

$$R_{m,\pm}(s) := G_{m,\pm}^{\text{corr}}(s) - G_{m,\pm}^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_{\pm})| \ll L_I^2 S_2, \quad |\delta q'(t_{\pm})| \ll L_I S_2, \quad |\delta q''(t_{\pm})| \ll S_2,$$

where L_I is the local window length, and that in the fixed-scaled regime

$$L_I \asymp \frac{x_0}{Q}, \quad |q_0(t_{\pm})| \asymp Q.$$

Then

$$\boxed{\|R_{m,\pm}(s)\| \ll S_2}$$

uniformly for $|s| < \rho_0/Q$.

Proof. We estimate the entries of $R_{m,\pm}(s)$ one by one.

(1,1)-entry. Since

$$R_{11} = \frac{1}{\pi} \cdot 2\delta q(t_{\pm}),$$

we have

$$|R_{11}| \ll |\delta q(t_{\pm})| \ll L_I^2 S_2 \ll \frac{S_2}{Q^2} \ll S_2.$$

(1,2)- and (2,1)-entries. Since

$$R_{12} = R_{21} = \frac{1}{\pi} \cdot \frac{1}{2}\delta q'(t_{\pm}),$$

we get

$$|R_{12}| = |R_{21}| \ll |\delta q'(t_{\pm})| \ll L_I S_2 \ll \frac{S_2}{Q} \ll S_2.$$

(2,2)-entry. We have

$$R_{22} = \frac{1}{12\pi} \left(\delta q''(t_{\pm}) + 2 \left((q_0 + \delta q)^3 - q_0^3 \right) (t_{\pm}) \right).$$

Expand the cubic difference:

$$(q_0 + \delta q)^3 - q_0^3 = 3q_0^2 \delta q + 3q_0 (\delta q)^2 + (\delta q)^3.$$

Hence

$$R_{22} = \frac{1}{12\pi} \left(\delta q'' + 6q_0^2 \delta q + 6q_0 (\delta q)^2 + 2(\delta q)^3 \right) (t_{\pm}).$$

Now estimate each term.

First,

$$|\delta q''(t_{\pm})| \ll S_2.$$

Second, using $|q_0(t_{\pm})| \asymp Q$ and $|\delta q(t_{\pm})| \ll L_I^2 S_2 \ll S_2/Q^2$,

$$|q_0(t_{\pm})^2 \delta q(t_{\pm})| \ll Q^2 \cdot \frac{S_2}{Q^2} = S_2.$$

Third,

$$|q_0(t_{\pm})(\delta q(t_{\pm}))^2| \ll Q \left(\frac{S_2}{Q^2} \right)^2 = \frac{S_2^2}{Q^3}.$$

Since from the tail-curvature theorem one already has $S_2 = o(Q)$, for Q sufficiently large this is $O(S_2/Q^2)$ and hence $\ll S_2$.

Finally,

$$|(\delta q(t_{\pm}))^3| \ll \left(\frac{S_2}{Q^2} \right)^3 = \frac{S_2^3}{Q^6} \ll S_2.$$

Therefore

$$|R_{22}| \ll S_2.$$

Collecting the four entrywise bounds, we obtain

$$|R_{11}|, |R_{12}|, |R_{21}|, |R_{22}| \ll S_2.$$

Since all norms on 2×2 matrices are equivalent,

$$\|R_{m,\pm}(s)\| \ll S_2,$$

uniformly on the microscopic disk. This proves the proposition. $\square$

Proposition 10.3 (Entry-by-entry perturbation estimate for the corrected mixed block). *Let*

$$q(t) = q_0(t) + \delta q(t), \quad \Phi(t) = \Phi_0(t) + \delta \Phi(t),$$

where q_0 is the baseline local-model-plus-affine-jet phase derivative and δq is the corrected remainder. Write

$$t_{\pm} = m \pm \frac{s}{2}, \quad \Delta(s) := \Phi(t_-) - \Phi(t_+), \quad \Delta_0(s) := \Phi_0(t_-) - \Phi_0(t_+),$$

and set

$$\delta \Delta(s) := \Delta(s) - \Delta_0(s).$$

Let $N_m^{\text{corr}}(s)$ be the corrected finite- s mixed block built from q, Δ , and let $N_m^{(0)}(s)$ be the corresponding block built from q_0, Δ_0 . Define

$$\tilde{R}_m(s) := N_m^{\text{corr}}(s) - N_m^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_{\pm})| \ll L_I^2 S_2, \quad |\delta q'(t_{\pm})| \ll L_I S_2, \quad |\delta q''(t_{\pm})| \ll S_2,$$

and

$$|\delta \Delta(s)| \ll L_I^3 S_2,$$

with

$$L_I \asymp \frac{x_0}{Q}, \quad |q_0(t_{\pm})| \asymp Q, \quad |s| \asymp L_I.$$

Then

$$\|\tilde{R}_m(s)\| \ll S_2$$

uniformly for $|s| < \rho_0/Q$.

Proof. Write the corrected mixed block entrywise as

$$N_m^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta}{s} & \frac{\sin \Delta + q(t_+)s \cos \Delta}{s^2} \\ -\frac{\sin \Delta + q(t_-)s \cos \Delta}{s^2} & \frac{(q(t_-) + q(t_+))s \cos \Delta + (2 - q(t_-)q(t_+)s^2) \sin \Delta}{2s^3} \end{pmatrix},$$

and analogously for $N_m^{(0)}(s)$ with (q_0, Δ_0) .

We estimate the perturbation entry by entry. Throughout, use

$$\sin(\Delta_0 + \delta\Delta) - \sin \Delta_0 = O(|\delta\Delta|), \quad \cos(\Delta_0 + \delta\Delta) - \cos \Delta_0 = O(|\delta\Delta|).$$

(1,1)-entry.

$$\tilde{R}_{11} = -\frac{2}{\pi s} (\sin \Delta - \sin \Delta_0),$$

hence

$$|\tilde{R}_{11}| \ll \frac{|\delta\Delta|}{|s|} \ll \frac{L_I^3 S_2}{L_I} \ll L_I^2 S_2 \ll S_2.$$

(1,2)-entry. The numerator perturbation is

$$(\sin \Delta - \sin \Delta_0) + (q(t_+) - q_0(t_+))s \cos \Delta + q_0(t_+)s(\cos \Delta - \cos \Delta_0).$$

Therefore

$$|\tilde{R}_{12}| \ll \frac{|\delta\Delta| + |\delta q(t_+)| |s| + |q_0(t_+)| |s| |\delta\Delta|}{|s|^2}.$$

Using

$$|\delta\Delta| \ll L_I^3 S_2, \quad |\delta q(t_+)| \ll L_I^2 S_2, \quad |q_0(t_+)| \asymp Q, \quad |s| \asymp L_I \asymp Q^{-1},$$

we get

$$|\tilde{R}_{12}| \ll \frac{L_I^3 S_2}{L_I^2} + \frac{L_I^3 S_2}{L_I^2} + \frac{Q L_I^4 S_2}{L_I^2} \ll \frac{S_2}{Q} \ll S_2.$$

The same argument gives

$$|\tilde{R}_{21}| \ll S_2.$$

(2,2)-entry. Write the numerator as

$$F(q_-, q_+, \Delta; s) := (q_- + q_+)s \cos \Delta + (2 - q_- q_+ s^2) \sin \Delta.$$

Then

$$\tilde{R}_{22} = \frac{F(q(t_-), q(t_+), \Delta; s) - F(q_0(t_-), q_0(t_+), \Delta_0; s)}{2\pi s^3}.$$

A first-order expansion of F in (q_-, q_+, Δ) gives

$$|F - F_0| \ll |s| |\delta q(t_-)| + |s| |\delta q(t_+)| + |s|^2 Q |\delta q(t_-)| + |s|^2 Q |\delta q(t_+)| + (|s|Q + 1) |\delta\Delta|.$$

Using

$$|\delta q(t_{\pm})| \ll L_I^2 S_2, \quad |\delta\Delta| \ll L_I^3 S_2, \quad |s| \asymp L_I \asymp Q^{-1},$$

we obtain

$$|F - F_0| \ll L_I^3 S_2.$$

Hence

$$|\tilde{R}_{22}| \ll \frac{L_I^3 S_2}{|s|^3} \ll S_2.$$

Collecting the four entrywise bounds and using equivalence of norms on 2×2 matrices gives

$$\|\tilde{R}_m(s)\| \ll S_2,$$

uniformly on the microscopic disk. $\square$

Proposition 10.4 (Corrected finite- s holomorphic whitening). *Assume:*

(1) *the corrected same-point and mixed blocks are holomorphic on*

$$|s| < \rho_0/Q;$$

(2) *the baseline blocks satisfy*

$$\lambda_{\min}(G_{m,\pm}^{(0)}(0)) \gg Q;$$

(3) *the perturbation satisfies*

$$\sup_{|s| < \rho_0/Q} \|R_{m,\pm}(s)\| = o(Q).$$

Then, after shrinking ρ_0 if necessary, the blocks

$$G_{m,\pm}^{\text{corr}}(s)$$

remain uniformly positive definite on $|s| < \rho_0/Q$, and the inverse square roots

$$G_{m,\pm}^{\text{corr}}(s)^{-1/2}$$

are holomorphic there. Consequently

$$\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m) = G_{m,-}^{\text{corr}}(s)^{-1/2} N_m^{\text{corr}}(s) G_{m,+}^{\text{corr}}(s)^{-1/2}$$

is holomorphic on $|s| < \rho_0/Q$.

Proof. By the explicit finite- s formulas, the only apparent poles at $s = 0$ arise from powers of $1/s$, $1/s^2$, $1/s^3$ in the mixed block. These are removable because $\Delta_m(s)$ is odd in s and hence

$$\sin \Delta_m(s) = O(s),$$

while the off-diagonal and $(2, 2)$ numerators vanish to order 2 and 3, respectively. The zero-induced singularities stay outside the microscopic disk by denominator comparability and the zero-free-region geometry.

By the baseline spectral gap and the perturbation bound,

$$\|G_{m,\pm}^{\text{corr}}(s) - G_{m,\pm}^{(0)}(0)\| \leq \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| + \|R_{m,\pm}(s)\|,$$

and the right-hand side is $o(Q) + O(\rho_0 Q)$. For ρ_0 small and m large, this is less than half the baseline spectral gap. Hence

$$\lambda_{\min}(G_{m,\pm}^{\text{corr}}(s)) \gg Q$$

uniformly on the disk. Holomorphic functional calculus then gives the holomorphic inverse square roots, and the whitened block is holomorphic as a product of holomorphic factors. $\square$

Proposition 10.5 (Corrected local deformation decomposition). *Let*

$$\Delta_{\zeta}(m, \sigma) := \widehat{\Omega}_{\zeta}^{\text{corr}}(s; m) - \widehat{\Omega}_{\text{bg}}^{\text{corr}}(s; m), \quad s = \sigma,$$

where $\widehat{\Omega}_{\text{bg}}^{\text{corr}}(s; m)$ denotes the corrected whitened block at background value-channel parameter $S(m) = 0$.

Assume:

(1) *the corrected same-point and mixed blocks are holomorphic on*

$$|s| < \rho_0/Q;$$

(2) *the same-point perturbation estimate of Proposition 10.2 and the mixed-block perturbation estimate of Proposition 10.3 hold;*

(3) *the same-point blocks remain uniformly positive definite on the microscopic disk.*

Then the corrected local deformation admits the expansion

$$\Delta_{\zeta}(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_{\zeta}(m, \sigma),$$

where $A_{\text{val}}(m, \sigma)$ is the pure value-channel derivative of the jet-limit block, and

$$\|R_{\zeta}(m, \sigma)\| \ll S(m)^2 + S_2 + \text{Tail}(m, \sigma) + \text{Err}(m, \sigma).$$

In particular, in any regime where

$$S_2 + \text{Tail} + \text{Err} \ll L(m)S(m)^2,$$

one has

$$\|R_\zeta(m, \sigma)\| \ll L(m)S(m)^2.$$

Proof. Consider the analytic map

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

By Proposition 10.4, this map is holomorphic on the open set where $G_\pm$ are uniformly positive definite.

Write the corrected blocks as

$$G_{m,\pm}^{\text{corr}}(s) = G_{m,\pm}^{\text{bg}}(s) + S(m) \dot{G}_{m,\pm}(s) + E_{m,\pm}(s),$$

$$N_m^{\text{corr}}(s) = N_m^{\text{bg}}(s) + S(m) \dot{N}_m(s) + \tilde{E}_m(s),$$

where the $\dot{G}_{m,\pm}, \dot{N}_m$ terms are the pure value-channel first variations and the error terms satisfy

$$\|E_{m,\pm}(s)\| + \|\tilde{E}_m(s)\| \ll S_2 + \text{Tail} + \text{Err}.$$

By Propositions 10.2 and 10.3, this error control is uniform on the microscopic disk.

Applying first-order Taylor expansion of $\mathcal{W}$ at the background point gives

$$\hat{\Omega}_\zeta^{\text{corr}}(s; m) = \hat{\Omega}_{\text{bg}}^{\text{corr}}(s; m) + S(m) D\mathcal{W}_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+] + R_\zeta(m, \sigma),$$

where the remainder satisfies

$$\|R_\zeta(m, \sigma)\| \ll S(m)^2 + \|E_{m,-}\| + \|\tilde{E}_m\| + \|E_{m,+}\|.$$

By definition of the pure value-channel derivative, the first-order coefficient is exactly $A_{\text{val}}(m, \sigma)$. This yields

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

with the claimed bound on R_ζ . □

11. BOUNDARY ESTIMATE AND ODD HOLOMORPHIC EXPANSION

Define the corrected transverse scalar

$$H_m(s) := \Phi_K(\hat{\Omega}_\zeta^{\text{corr}}(s; m)).$$

Proposition 11.1 (Boundary estimate). *On the microscopic boundary*

$$|s| = \rho_0/Q,$$

one has

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3}.$$

Proof. Write the corrected local deformation in the form

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma)$$

as in Proposition 10.5. Applying Φ_K and using the jet-limit value-channel vanishing

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0,$$

we obtain

$$H_m(s) = \Phi_K(\hat{\Omega}_\zeta^{\text{corr}}(s; m)) = \Phi_K(R_\zeta(m, \sigma)).$$

We now split the remainder into its three natural pieces,

$$R_\zeta = \Omega_{\text{curv}} + \Omega_{\text{tail}} + \Omega_{\text{err}}.$$

Curvature contribution. By the local curvature estimate and the corrected transfer theorem, the curvature channel has the form

$$\Omega_{\text{curv}}(s; m) = C_m(s) V(s),$$

where

$$|C_m(s)| \ll L(m)S(m)^2$$

on the microscopic boundary, and the explicit transverse profile satisfies

$$|\Phi_K(V(s))| \ll_{\rho_0} Q^{-3} \quad (|s| = \rho_0/Q).$$

Therefore

$$|\Phi_K(\Omega_{\text{curv}}(s; m))| \leq |C_m(s)| |\Phi_K(V(s))| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3}.$$

Tail and error contributions. By Proposition 10.1, the omitted smooth part is holomorphic on the microscopic disk, and by Propositions 10.2 and 10.3 its contribution to the corrected finite- s blocks is perturbative. The same holds for the estimation term. After whitening, these perturbations remain of the same order up to bounded operator factors, so on $|s| = \rho_0/Q$,

$$\|\Omega_{\text{tail}}(s; m)\| + \|\Omega_{\text{err}}(s; m)\| \ll \frac{\text{Tail}(m, \sigma) + \text{Err}(m, \sigma)}{Q^3}.$$

Applying the bounded linear functional Φ_K gives

$$|\Phi_K(\Omega_{\text{tail}}(s; m))| + |\Phi_K(\Omega_{\text{err}}(s; m))| \ll \frac{\text{Tail}(m, \sigma) + \text{Err}(m, \sigma)}{Q^3}.$$

Combining the three pieces yields

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2 + \text{Tail}(m, \sigma) + \text{Err}(m, \sigma)}{Q^3}.$$

In the corrected regime where

$$\text{Tail}(m, \sigma) + \text{Err}(m, \sigma) \ll L(m)S(m)^2,$$

this simplifies to

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3},$$

as claimed. $\square$

Proposition 11.2 (Odd microscopic expansion). *The function $H_m(s)$ is odd and holomorphic on $|s| < \rho_0/Q$, and therefore admits an expansion*

$$H_m(z/Q^2) = \sum_{r \geq 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}}, \quad |z| < \rho_0,$$

with

$$|c_{2r+1}(m)| \ll_{\rho_0} L(m)S(m)^2 \rho_0^{-(2r+1)}.$$

Proof. Oddness follows from symmetric placement and the jet-basis swap symmetry:

$$\hat{\Omega}_\zeta^{\text{corr}}(-s; m) = J_0 \hat{\Omega}_\zeta^{\text{corr}}(s; m) J_0,$$

with

$$\Phi_K(J_0 X J_0) = -\Phi_K(X).$$

Holomorphy follows from Proposition 10.4, and the coefficient bounds are immediate from Cauchy's estimate applied to Proposition 11.1. $\square$

12. CALIBRATED N -POINT CONTRADICTION

For $N \leq \rho_0 Q$, define the arithmetic microscopic nodes

$$s_j = \frac{j}{Q^2}, \quad j = 1, \dots, N,$$

and let $a_j^{(N)}$ be the explicit odd-moment-canceling coefficients. Once and for all, normalize the odd-moment-canceling coefficients so that

$$\sum_{j=1}^N a_j^{(N)} = 1.$$

This does not affect the vanishing of the first $N - 1$ odd moments.

Define the corrected N -point zeta transverse scalar

$$\Xi_\zeta^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_m(s_j) = \sum_{j=1}^N a_j^{(N)} \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s_j; m)).$$

Proposition 12.1 (Corrected N -point transverse estimate for the full corrected scalar). *Let*

$$H_m(s) := \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m))$$

be the full corrected transverse scalar. For $N = \lfloor \alpha Q \rfloor$ with $\alpha > 0$ sufficiently small, define

$$s_j = \frac{j}{Q^2}, \quad \Xi_\zeta^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_m(s_j).$$

Then

$$|\Xi_\zeta^{(N)}(m)| \ll L(m) S(m)^2 e^{-\delta Q}$$

for some $\delta > 0$.

Proof. By Proposition 11.2, the full corrected scalar $H_m(s)$ is odd and holomorphic on $|s| < \rho_0/Q$ and admits the expansion

$$H_m(z/Q^2) = \sum_{r \geq 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}},$$

with

$$|c_{2r+1}(m)| \ll L(m) S(m)^2 \rho_0^{-(2r+1)}.$$

Applying the standard odd-moment cancellation with the coefficients $a_j^{(N)}$ kills the first $N - 1$ odd terms of this full corrected expansion. The surviving moment factor is exponentially tame for $N = \alpha Q$, which gives the stated bound. $\square$

Lemma 12.2 (Crude polynomial bound for the local value scale). *Let*

$$S(m) = \sum_{\rho} f_{\beta_\rho, \gamma_\rho}(m),$$

where

$$f_{\beta, \gamma}(m) = \frac{1 - \beta}{(1 - \beta)^2 + (2m - \gamma)^2} + \frac{\beta}{\beta^2 + (2m - \gamma)^2}.$$

Then

$$S(m) \ll Q^2, \quad Q \asymp \log |m|.$$

Consequently

$$L(m) \ll Q$$

and

$$B_\zeta(m) L(m) S(m) \ll Q^4.$$

Proof. By the zero-free region, for zeros at relevant height one has

$$\min(\beta, 1 - \beta) \gg \frac{1}{Q}.$$

Hence

$$f_{\beta, \gamma}(m) \ll \frac{Q}{1 + (2m - \gamma)^2}.$$

Partition the zeros into unit shells

$$\mathcal{S}_k := \{\rho : k \leq |2m - \gamma_\rho| < k + 1\}, \quad k = 0, 1, 2, \dots$$

Standard zero counting gives

$$\#\mathcal{S}_k \ll Q$$

for each unit shell near height m . Therefore

$$S(m) \ll \sum_{k \geq 0} \#\mathcal{S}_k \frac{Q}{1 + k^2} \ll Q \sum_{k \geq 0} \frac{Q}{1 + k^2} \ll Q^2.$$

Since

$$L(m) = \log\left(3 + |m| + \frac{1}{S(m)}\right),$$

we trivially have $L(m) \ll Q$, and the last estimate follows from $B_\zeta(m) \asymp Q$. □

Proposition 12.3 (Toy N -point preservation). *Let*

$$\Xi_{\text{toy}}^{(N)}(u, d) := \sum_{j=1}^N a_j^{(N)} \Phi_K(\Delta_{\text{toy}}(u, d; s_j)), \quad s_j = \frac{j}{Q^2},$$

with the normalization

$$\sum_{j=1}^N a_j^{(N)} = 1.$$

Assume the toy local deformation satisfies, uniformly in the microscopic nodes,

$$\Delta_{\text{toy}}(u, d; s_j) = u^2 M(d) + u^4 E_j(u, d),$$

where

$$\sup_{1 \leq j \leq N} \|E_j(u, d)\| \ll 1$$

for d in compact subsets of $(0, \infty)$. Then

$$\boxed{\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O\left(u^4 \sum_{j=1}^N |a_j^{(N)}|\right)}.$$

In particular, if

$$\sum_{j=1}^N |a_j^{(N)}| \ll 1$$

or more generally if $u^2 \sum_j |a_j^{(N)}| \rightarrow 0$, then

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

and therefore

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2$$

uniformly on compact d -ranges.

Proof. Apply Φ_K to the toy expansion:

$$\Phi_K(\Delta_{\text{toy}}(u, d; s_j)) = u^2 \Phi_K(M(d)) + u^4 \Phi_K(E_j(u, d)).$$

Summing over j and using

$$\sum_{j=1}^N a_j^{(N)} = 1$$

gives

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + u^4 \sum_{j=1}^N a_j^{(N)} \Phi_K(E_j(u, d)).$$

Since Φ_K is bounded and $\sup_j \|E_j(u, d)\| \ll 1$, the error term is

$$O\left(u^4 \sum_{j=1}^N |a_j^{(N)}|\right).$$

This proves the claim. $\square$

Theorem 12.4 (Final calibrated N -point contradiction). *Assume the calibration theorem, the corrected finite- s holomorphic whitening theorem, and the corrected N -point transverse estimate above.*

Then no fixed off-line toy anomaly can occur. Consequently the Riemann Hypothesis holds.

Proof. The calibration theorem gives, in the fixed small- x_0 regime,

$$u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

On the toy side, Proposition 12.3 gives

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O\left(u^4 \sum_{j=1}^N |a_j^{(N)}|\right).$$

Under the chosen normalization and coefficient bounds, this is

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

and since $\Phi_K(M(d)) \neq 0$ on every compact d -interval,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

On the zeta side, Proposition 12.1 gives

$$|\Xi_\zeta^{(N)}(m)| \ll L(m) S(m)^2 e^{-\delta Q}.$$

It remains only to compare these sizes. By Lemma 12.2, one has

$$S(m) \ll Q^2, \quad L(m) \ll Q, \quad B_\zeta(m) L(m) S(m) \ll Q^4.$$

Therefore

$$\frac{L(m) S(m)^2 e^{-\delta Q}}{(x_0/B_\zeta(m)) S(m)} = \frac{B_\zeta(m)}{x_0} L(m) S(m) e^{-\delta Q} \ll Q^4 e^{-\delta Q} \rightarrow 0.$$

Hence for Q sufficiently large,

$$|\Xi_\zeta^{(N)}(m)| < |\Xi_{\text{toy}}^{(N)}(u, d)|.$$

This is impossible if the corrected local zeta deformation realizes the local off-line toy anomaly with parameter $d \in D$, because the calibrated N -point construction was set up precisely so that the toy model is the jet-limit local avatar of an off-line zero configuration. Thus no off-line zero can occur. Since every violation of the Riemann Hypothesis would produce such an off-line local anomaly, all nontrivial zeros must lie on the critical line. Hence RH holds. $\square$